

	STUDENT NAME			COURSE NAME	Certified In CAE - ANSYS
	REGISTER. NO			TOTAL HOURS	120 HRS
	MOBILE.NO			JOINING DATE	
	CLASS TIMING			STAFF NAME	

SESSION	DATE & HOURS	MODULES	HOURS	STUDENT SIGN	STAFF SIGN
01 SESSION	DATE :	ANSYS Introduction to FEA & Interface - What is FEA, CAE, and simulation? - Engineering Facts, purpose of FEA(safety, deformation checks) - Discretization: Nodes, Elements, - Degrees of Freedom (DOF) - FOS manifestation determining - FEA workflow & types of analysis - ANSYS Workbench interface, unit systems, and project setup - Real component selection (bracket, beam, frame) and basic loads (force, pressure, gravity)	90 Mins		
	IN TIME :				
	OUT TIME :				
02 SESSION	DATE :	Engineering Data (Materials) - What is Engineering Data? - Material models overview (linear elastic, stress-strain curves) - Stiffness, Strength, Weight, Plastic behaviour - Isotropic vs orthotropic materials - Key properties: Young's modulus(E), Poisson's ratio (v), yield strength - Temperature-dependent properties - Importing, editing, and creating custom materials	90 Mins		
	IN TIME :				
	OUT TIME :				
03 SESSION	DATE :	Geometry Preparation - CAD cleanup for FEA - Importing geometry - Solid vs Surface vs Line bodies - Suppressing features - Share topology - Geometry defeaturing techniques - Fillet / chamfer removal - Hole removal / filling - Gap & overlap repair - Gap & overlap repair - Face merging - Body stitching / healing - Symmetry modelling - Merge / combine bodies - Boolean operations - Contact preparation - Named selections - Unit verification	60 Mins		
	IN TIME :				
	OUT TIME :				

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04 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	PRACTICE GEOMETRY PREPARATION Practice geometry preparation as per the requirement and conditions.	510 Mins		
05 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	Meshing Fundamentals - Purpose of meshing - Element types (1D, 2D, 3D) - Tetra vs Hex elements - Mesh controls & sizing - Relevance & advanced sizing - Inflation layers - Mesh quality checks	60 Mins		
06 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	PRACTICE MESHING AND GEOMETRY PREPARATION Practice the meshing and geometry preparation methods as per the requirement	2400 Mins		
07 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	Boundary Conditions & Loads - Boundary condition fundamentals - Rigid body motion prevention - Fixed support - Displacement constraint - Frictionless support - Symmetry boundary condition - Force load - Pressure load - Moment load - Remote force - Gravity load - Thermal load	90 Mins		

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08 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	Contacts & Connections • Contact fundamentals • Contact vs target surfaces • Bonded contact • Frictionless contact • Frictional contact • No separation contact • Penetration & gap treatment • Contact detection methods	60 Mins		
09 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	Static Structural Analysis • Linear static analysis • Stress, strain & deformation results • Factor of safety • Reaction forces • Result interpretation • Engineering validation	60 Mins		
10 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	Linear Analysis Concepts • Assumptions of linearity • Linear material behaviour • Small deformation theory • Applicability & limitations • Industrial examples	60 Mins		
11 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	Non-Linear Analysis • Sources of nonlinearity • Material nonlinearity • Geometric nonlinearity • Contact nonlinearity • Large deformation analysis • Convergence controls	60 Mins		
12 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	Dynamic Analysis • Transient dynamic analysis • Time-dependent loading • Damping concepts • Frequency response plots	60 Mins		

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13 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	Thermal Analysis - Steady-state thermal analysis - Transient thermal analysis - Heat conduction, convection, radiation - Thermal boundary conditions - Thermal stress coupling	60 Mins		
14 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	PRACTICE ANALYSIS CONCEPTS Practice the analysis concepts as per the conditions and requirement	3000 Mins		
15 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	Bolted & Welded Joints - Bolt pretension - Beam vs solid bolts - Contact in bolted joints - Weld modelling techniques - Stress concentration in joints	90 Mins		
16 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	Post-Processing & Reporting - Stress plots & deformation plots - Path & probe results - Result validation checks - Image & plot export - Industrial report format	90 Mins		
17 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	PROJECTS INDUSTRIAL CASE STUDIES - Machine component analysis - Bracket & support structure - Pressure component - Thermal-structural case - Failure interpretation - Interview-oriented discussion	300 Mins		

COURSE **COMPLETED**

Staff sign

Student sign